

CS 4032 – Web Programming



CSS - Cascading Style Sheets



History of CSS

- 1994: Cascading HTML style sheets—a proposal
 - Hakon W Lie proposed CSS
 - Working with Tim-Berners Lee at CERN
- 1996: CSS1 standard, recommended by W3C
 - Defines basic styling elements like font, color, alignment, margin, padding, etc.
- 1998: CSS2 standard, recommended by W3C
 - Adds positioning schemes, z-index, new font properties, etc.
- 2011: CSS3 standards, structured in modules
 - Add more powerful selectors, responsive web design, etc.



What is CSS ?

- CSS defines how to display an html document
 - fonts, colors, layouts, etc.
- Usually CSS styles are specified in an external file
- Separates visual presentation (CSS) from document structure (HTML)
- Enables developers to change the layout and look of all documents in a website by just editing a single file



Need for CSS

Not much flexibility in positioning of content

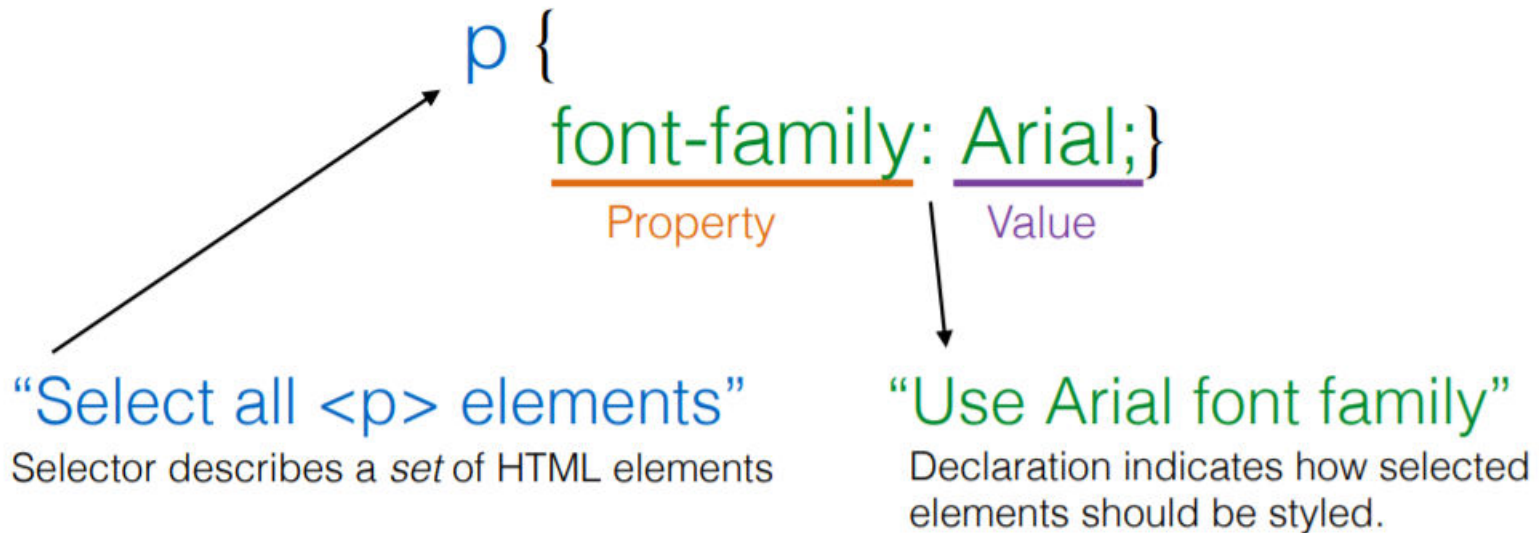
Inefficient and repetitive

```
<body>
<table width="100%" height="60" border="0" cellspacing="5" cellpadding="5" bgcolor="#000000">
  <tr>
    <td align="left"><table width="547" border="0" align="center" cellpadding="1" cellspacing="1">
      <tr>
        <td width="96"><font face="Verdana, Geneva, sans-serif" color="white"><h3>Home</h3></font></td>
        <td width="113"><font face="Verdana, Geneva, sans-serif" color="white"><h3>About</h3></font></td>
        <td width="128"><font face="Verdana, Geneva, sans-serif" color="white"><h3>Services</h3></font></td>
        <td width="114"><font face="Verdana, Geneva, sans-serif" color="white"><h3>Clients</h3></font></td>
        <td width="80"><font face="Verdana, Geneva, sans-serif" color="white"><h3>Contact</h3></font></td>
      </tr>
    </table></td>
  </tr>
</table>
```

Cannot stack multiple objects (e.g., images) on top of one another

CSS Syntax

- selector { property: value; property: value; ... }



CSS Selectors

CSS Selector	CSS	HTML
Tag name	<pre>h1 { color: red; }</pre>	<pre><h1>Today's Specials</h1></pre>
Class attribute	<pre>.large { font-size: 16pt; }</pre>	<pre><p class="large">...</pre>
Tag and Class	<pre>p.large {...}</pre>	<pre><p class="large">...</pre>
Element id	<pre>#p20 { font-weight: bold; }</pre>	<pre><p id="p20">...</pre>

CSS Type Selectors

```
h2, h3 {  
    color: LightGreen;  
}
```

“Select all <h2> and <h3> elements”

Type selector selects one or more element types.

```
* {  
    color: LightGreen;  
}
```

“Select all elements”

Universal selector selects all elements.

CSS Class Selectors

```

```

“Label element with imageLarge class”

```
.imageLarge {  
    width: 200px;  
    height: 200px;  
}
```

“Define class imageLarge.”

```

```

```
img.large {  
    width: 200px;  
    height: 200px;  
}
```

“Define large class that applies only to elements”

```
.transparent {  
    opacity: .50;  
}
```

“Define transparent class”

CSS Id Selectors

```
<div id="exampleElem">  
  Some text  
</div>
```

```
#exampleElem {  
  font-weight: bold;  
}
```

Some text

Additional Selector Types

Selector	Meaning	Example	
<i>Descendant selector</i>	Matches all descendants of an element	<code>p a { }</code>	Select <a> elements inside <p> elements
<i>Child selector</i>	Matches a direct child of an element	<code>h1>a { }</code>	Select <a> elements that are directly contained by <h1> elements.
<i>First child selector</i>	Matches the first child of an element	<code>h1:first-child { }</code>	Select the the elements that are the first child of a <h1> element.
<i>Adjacent selector</i>	Matches selector	<code>h1+p { }</code>	Selects the first <p> element after any <h1> element
<i>Negation selector</i>	Selects all elements that are not selected.	<code>body *:not(p)</code>	Select all elements in the body that are not <p> elements.
<i>Attribute selector</i>	Selects all elements that define a specific attribute.	<code>input[invalid]</code>	Select all <input> elements that have the invalid attribute.
<i>Equality attribute selector</i>	Select all elements with a specific attribute value	<code>p[class="invisible"]</code>	Select all <p> elements that have the invisible class.

Pseudo Selectors/Classes

Syntax:

selector:pseudo-class {property: value; ...}

selector.class:pseudo-class {property: value; ...}

```
.invisible {  
  display: none;  
}
```

```
input:invalid {  
  border: 2px solid red;  
}
```

```
input:invalid + div {  
  display: block;  
}
```

```
input:focus + div {  
  display: none;  
}
```

```
<label>  
  Email: <input type="email" />  
  <div class="invisible">Please enter a valid email.</div>  
</label>
```

Email:

“Select elements with the invalid attribute.”

“Select elements that have focus.”

Examples of pseudo classes

- :active - elements activated by user. For mouse clicks, occurs between mouse down and mouse up.
- :checked - radio, checkbox, option elements that are checked by user
- :disabled - elements that can't receive focus
- :empty - elements with no children
- :focus - element that currently has the focus
- :hover - elements that are currently hovered over by mouse
- :invalid - elements that are currently invalid
- :link - link element that has not yet been visited
- :visited - link element that has been visited
- :lang: adds a style to an element with a specific lang attribute

https://www.w3schools.com/css/tryit.asp?filename=trycss_pseudo-class_links



Pseudo-elements

- syntax:
 - selector:pseudo-element {property: value; ...}
 - selector.class:pseudo-element {property: value; ...}
- list of pseudo-elements:
 - :first-letter – adds style to the first character of a text
 - p:first-letter { font-size: xx-large; }
 - https://www.w3schools.com/css/tryit.asp?filename=trycss_pseudo-element
 - :first-line – adds style to the first line of a text
 - p:first-line { color: red }
 - :before – adds content before an element
 - h1:before { content: url(smiley.jpg)}
 - https://www.w3schools.com/css/tryit.asp?filename=trycss_before
 - :after – adds content after an element
 - h1:after { content: url(smiley.jpg)}

A simple CSS example

```
body {background-color: #aabbcc}
```

```
h1 {font-style: italic; font-size: 36pt}
```

```
h2 {color: blue}
```

```
p {margin-left: 50px}
```

- It specifies that:
 - the <body> tag should have a background color of #aabbcc
 - the <h1> heading tag should use an italic font of size 36pt
 - the <h2> heading tag should use a blue color for displaying text
 - a paragraph <p> tag should start at an offset of 50 pixels on the left

Adding style sheets to a document

- External style sheet:

```
<head>
```

```
  <link rel="stylesheet" type="text/css" href="style.css">
```

```
</head>
```

- Internal style sheet:

```
<head>
```

```
  <style type="text/css">
```

```
    p { margin-left: 10px }
```

```
  </style>
```

```
</head>
```

- Inline style sheet:

```
<p style="color: red; margin-left: 10px">
```


Multiple style sheets

- If multiple style sheets are defined, their priority is:
 - 1) Inline style
 - 2) External style sheet
 - 3) Internal style sheet
 - 4) Browser default

Background properties

Property	Description
background	Sets all the background properties in one declaration
background-attachment	Sets whether a background image is fixed or scrolls with the rest of the page
background-color	Sets the background color of an element
background-image	Sets the background image for an element
background-position	Sets the starting position of a background image
background-repeat	Sets how a background image will be repeated

Background CSS examples

```
body {  
  background-color: blue;  
}  
h1 {  
  background-color: purple;  
}
```



```
body {  
  background-image: url("barn.jpg");  
}
```



Background CSS examples

```
body {  
  background-image: url("barn.jpg");  
  background-repeat: repeat-y;  
}
```



```
body {  
  background-image: url("barn.jpg");  
  background-repeat: no-repeat;  
}
```



Background CSS examples

```
body {  
  background-image: url("barn.jpg");  
  background-repeat: no-repeat;  
  background-position: right top;  
}
```

This image is positioned to the top right of this page.



```
body {  
  background-image: url("barn.jpg");  
  background-repeat: no-repeat;  
  background-position: right top;  
  background-attachment: fixed;  
}
```

This image is positioned to the top right of this page.

This image is positioned to the top right of this page.

When the page is scrolled the image will stay in place.



Text properties

Property	Description
color	Sets the color of text
direction	Specifies the text direction/writing direction
letter-spacing	Increases or decreases the space between characters in a text
line-height	Sets the line height
text-align	Specifies the horizontal alignment of text
text-decoration	Specifies the decoration added to text
text-indent	Specifies the indentation of the first line in a text-block
text-shadow	Specifies the shadow effect added to text
text-transform	Controls the capitalization of text
vertical-align	Sets the vertical alignment of an element
white-space	Specifies how white-space inside an element is handled
word-spacing	Increases or decreases the space between words in a text

Text CSS example

```
h2 {  
  text-transform: uppercase;  
  text-decoration: underline;  
  letter-spacing: 0.2em;  
  text-align: center;  
  line-height: 2em;  
  vertical-align: middle;  
  text-shadow: 1px 1px 0 #666666;  
}
```

THIS TEXT IS IMPORTANT

Display and visibility

- Hiding an element with consuming space:
visibility: hidden;
- Hiding an element without consuming space:
display: none;
- HTML elements are displayed as:
 - *block* – element takes the full width available and has a line break before and after it (e.g. h1, p, div)
 - *inline* – element takes as much width as necessary and does not force line breaks (e.g. a, span)
- Set the display mode with the *display* property (e.g. display: inline; display: block;)
- Make an image transparent:
opacity: x – on Firefox, x between 0.0 and 1.0
filter:alpha(opacity=x) – on IE, x between 0 and 100.

Visibility and layout

- Can force elements to be inline or block element.
 - display: inline
 - display: block
- Can cause element to not be laid out or take up any space
 - display: none
 - Very useful for content that is dynamically added and removed.
- Can cause boxes to be invisible, but
 - still take up space
 - visibility: hidden;

```
<ul>  
  <li>Home</li>  
  <li>Products</li>  
  <li class="coming-soon">Services</li>  
  <li>About</li>  
  <li>Contact</li>  
</ul>
```

```
li {  
  display: inline;  
  margin-right: 10px; }  
li.coming-soon {  
  display: none; }
```

Home Products About Contact

```
li {  
  display: inline;  
  margin-right: 10px; }  
li.coming-soon {  
  visibility: hidden; }
```

Home Products

About Contact

Font properties

Property	Description
font	Sets all the font properties in one declaration
font-family	Specifies the font family for text
font-size	Specifies the font size of text
font-style	Specifies the font style for text
font-variant	Specifies if a text should be displayed in a small-caps font
font-weight	Specifies the weight of the font

Font CSS example

```
p, div {  
    font-family: "Times New Roman", Arial, Serif;  
    font-style: italic;  
    font-size: 12px;  
}
```

List properties

Property	Description
list-style	Sets all the properties for a list in one declaration
list-style-image	Specifies an image as the list-item marker
list-style-position	Specifies where to place the list-item marker
list-style-type	Specifies the type of list-item marker

List CSS example

```
ul {  
    list-style-type: circle;  
    list-style-position: inside;  
}
```

Table properties

Property	Description
border-collapse	Specifies whether or not table borders should be collapsed
border-spacing	Specifies the distance between the borders of adjacent cells
caption-side	Specifies the placement of a table caption
empty-cells	Specifies whether or not to display borders and background on empty cells in a table
table-layout	Sets the layout algorithm to be used for a table

The CSS box model

- Each HTML element is a “box” from CSS perspective; it has a margin, border, padding and content



The CSS Box model (2)

- Margin: clears an area around the border; does not have a background color, is completely transparent
- Border: lies around content and padding; has the background color of the box
- Padding: clears an area around the content; has the background color of the box
- Content: the actual content (text) of the box
- The *height* and *width* attributes do not specify the total height and width of the HTML element; they are calculated by:

Total width=width+left padding+right padding+left border+
right border+left margin+right margin

Total height=height+top padding+bottom padding+top border+ bottom
border+top margin+bottom margin

Dimension properties

Property	Description
height	Sets the height of an element
max-height	Sets the maximum height of an element
max-width	Sets the maximum width of an element
min-height	Sets the minimum height of an element
min-width	Sets the minimum width of an element
width	Sets the width of an element

Margin and padding properties

Property	Description
margin	Sets all the margin properties in one declaration
margin-bottom	Sets the bottom margin of an element
margin-left	Sets the left margin of an element
margin-right	Sets the right margin of an element
margin-top	Sets the top margin of an element
padding	Sets all the padding properties in one declaration
padding-bottom	Sets the bottom padding of an element
padding-left	Sets the left padding of an element
padding-right	Sets the right padding of an element
padding-top	Sets the top padding of an element

Margin CSS example

```
div {  
    margin-top: 100px;  
    margin-bottom: 100px;  
    margin-left: 5%;  
    margin-right: 5%;  
}
```

Border and outline properties

Property	Description
border	Sets all the border properties in one declaration
border-bottom	Sets all the bottom border properties in one declaration
border-bottom-color	Sets the color of the bottom border
border-bottom-style	Sets the style of the bottom border
border-bottom-width	Sets the width of the bottom border
border-color	Sets the color of the four borders
border-left	Sets all the left border properties in one declaration
border-left-color	Sets the color of the left border
border-left-style	Sets the style of the left border
border-left-width	Sets the width of the left border
border-right	Sets all the right border properties in one declaration
border-right-color	Sets the color of the right border
border-right-style	Sets the style of the right border
border-right-width	Sets the width of the right border
border-style	Sets the style of the four borders
border-top	Sets all the top border properties in one declaration
border-top-color	Sets the color of the top border
border-top-style	Sets the style of the top border
border-top-width	Sets the width of the top border
border-width	Sets the width of the four borders
outline	Sets all the outline properties in one declaration
outline-color	Sets the color of an outline
outline-style	Sets the style of an outline
outline-width	Sets the width of an outline



Border CSS example

```
div {  
    border-top-style: solid;  
    border-bottom-style: solid;  
    border-left-style: dotted;  
    border-right-style: dotted;  
    border-width: 2px;  
    border-color: #00ff00;  
}
```

Positioning

- static positioning: the default mode; a static element is positioned according to the normal flow of the page;
- fixed positioning: element will not move even if the window is scrolled (position: fixed; top: 20px; right: 40px); elements are removed from the normal flow and can overlap other elements
- relative positioning: element is positioned relative to its normal position; element can be moved and overlap other elements , but the reserved space form the element is preserved in the normal flow (position: relative)
- absolute positioning: element is positioned relative to the first parent element that has a position other then static (if none, <html>); elements are removed from the normal flow and can overlap other elements; (position: absolute)
- z-index property defines the stack order of an element

Floating

- With floating, an html element can be pushed left or right allowing other elements to wrap around it
- HTML elements can float horizontally left or right (not vertically)
- Configured using the float: left | right | none | inherit display property



Positioning properties

Property	Description
bottom	Sets the bottom margin edge for a positioned box
clear	Specifies which sides of an element where other floating elements are not allowed
clip	Clips an absolutely positioned element
cursor	Specifies the type of cursor to be displayed
display	Specifies the type of box an element should generate
float	Specifies whether or not a box should float
left	Sets the left margin edge for a positioned box
overflow	Specifies what happens if content overflows an element's box
position	Specifies the type of positioning for an element
right	Sets the right margin edge for a positioned box
top	Sets the top margin edge for a positioned box
visibility	Specifies whether or not an element is visible
z-index	Sets the stack order of an element

Positioning Examples

Normal flow (default)

Lorem Ipsum

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Relative positioning

Lorem Ipsum

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```
p.example {  
  position: relative;  
  top: 10px;  
  left: 100px;  
}
```

Absolute positioning

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```
h3 {  
  position: absolute;  
  background-color: LightGray;  
  left: 350px;  
  width: 250px;  
}
```

Fixed positioning

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Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

```
h3 {  
  position: fixed;  
  background-color: LightGray;  
  left: 40px;  
  width: 250px;  
}
```

Floating elements

Lorem Ipsum

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```
h3 {  
  float: left;  
  background-color: LightGray;  
  left: 40px;  
  width: 250px;  
}
```



Stacking elements

- Elements taken out of normal flow may be stacked on top of each other
- Can set order with z-index property
 - Higher numbers appear in front
- Can set opacity of element, making occluded elements partially visible

```
h3 {  
  position: absolute;  
  background: LightGray;  
  opacity: 0.6;  
  z-index: 10;  
}
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do
~~Lorem ipsum~~ incididunt ut labore et dolore magna aliqua.

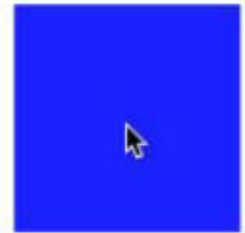
Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris
nisi ut aliquip ex ea commodo consequat.

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dolore eu fugiat nulla pariatur.

Transitions

- transition: [property time], ..., [property time]
 - When new class is applied, specifies the time it will take for each property to change
 - Can use all to select all changed properties

```
.box {  
  width: 100px;  
  height: 100px;  
  background-color: #0000FF;  
  transition: width 2s, height 2s, background-color 2s, transform 2s;  
}  
  
.box:hover {  
  background-color: #FFCCCC;  
  width: 200px;  
  height: 200px;  
  transform: rotate(180deg);  
}  
  
<div class="box"></div>
```



Cursor

- Can change the default cursor with cursor attribute
 - auto, crosshair, pointer, move, text, wait, help, url("cursor.gif")
- Should only do this if action being taken clearly matches cursor type

```
<a class="movableItem">Walt Whitman</a>
```

Walt  Whitman

```
a.movableItem {  
  cursor: move;  
}
```

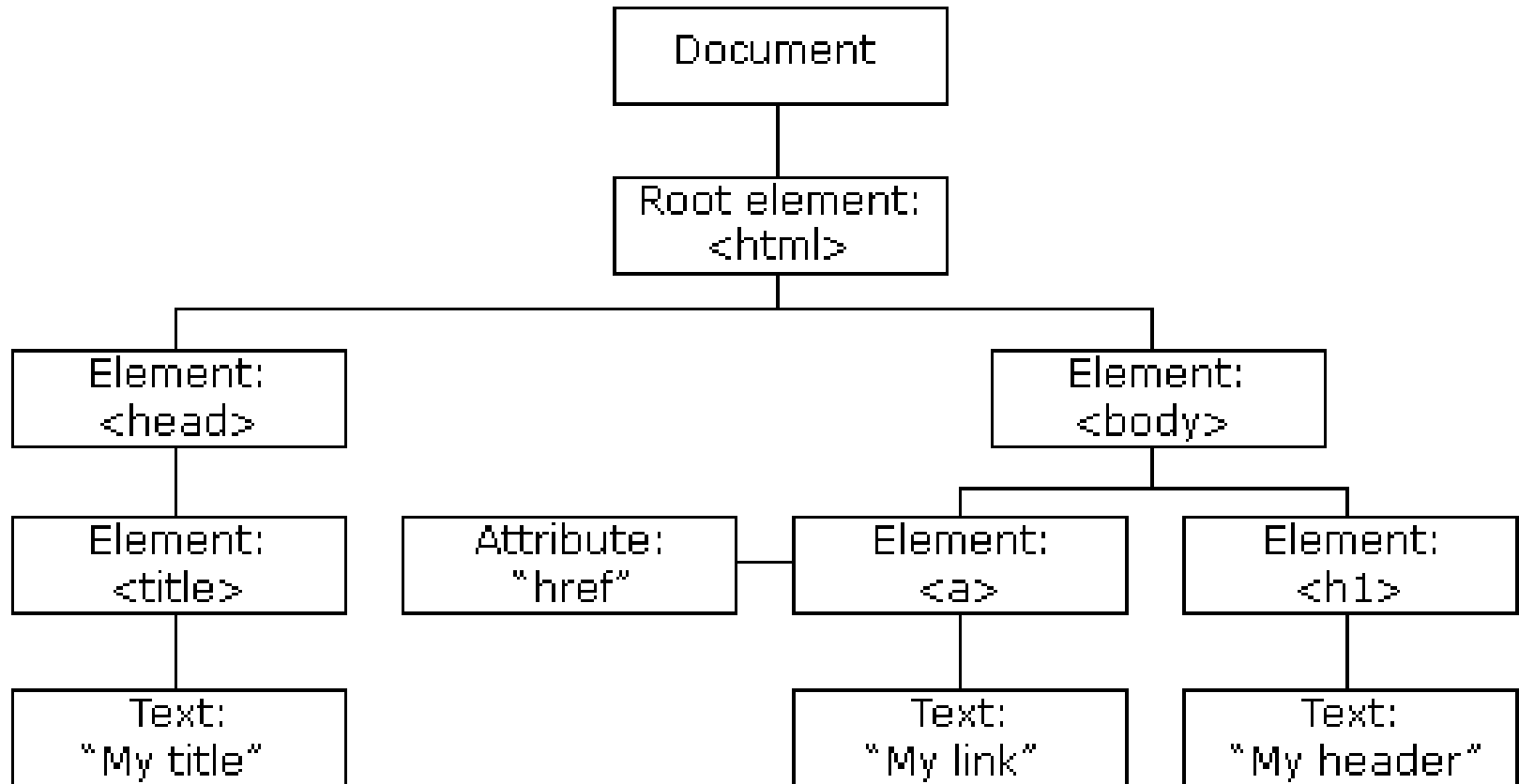
CSS Cursors

 auto	 move	 no-drop	 col-resize
 all-scroll	 pointer	 not-allowed	 row-resize
 crosshair	 progress	 e-resize	 ne-resize
 default	 text	 n-resize	 nw-resize
 help	 vertical-text	 s-resize	 se-resize
 inherit	 wait	 w-resize	 sw-resize

CSS level 3



DOM Tree



CSS Selectors (1)

- E[attr^="str"] - an E element whose "attr" attribute begins with "str"
 - E.g., a[href^='http://sales.'){color: teal;}
- E[attr\$="str"] - an E element whose "attr" attribute ends with "str"
 - E.g., a[href\$='.jsp']{color: purple;}
- E[attr*="str"] - an E element whose "attr" attribute contains substring "str"
 - E.g., img[src*='artwork']{
 border-color: #C3B087 #FFF #FFF #C3B087;
 }

CSS Selectors (2)

- `E:nth-child(n)` - an E element, the n-th child of its parent
- `E:nth-last-child(n)` - an E element, the n-th child of its parent, counting from the last one
- `E:nth-of-type(n)` - an E element, the n-th sibling of its type
- `E:nth-last-of-type(n)` - an E element, the n-th sibling of its type, counting from the last one
- `E:first-child` - an E element, first child of its parent (in CSS2.1)
- `E:last-child` - an E element, last child of its parent

CSS Selectors (3)

- E:first-of-type - an E element, first sibling of its type
- E:last-of-type - an E element, last sibling of its type
- E:only-child - an E element, only child of its parent
- E:only-of-type - an E element, only sibling of its type
- E:empty - an E element that has no children
- E:not(s) - an E element that does not match simple selector s
- E > F - an F element child of an E element (in CSS2.1)
- E + F - an F element immediately preceded by an E element (in CSS2.1)
- E ~ F - an F element preceded by an E element

Responsive web design

- Document should look good on any device (desktop, tablet, phone)
- Use CSS to shrink, enlarge, hide or move html content in order to look good on any screen
- Setting viewport:
 - `<meta name="viewport" content="width=device-width, initial-scale=1.0">`
 - sets the width of the page to follow the screen width of the device
 - sets the initial zoom level to 1 (no zoom)
- Do not rely on a particular viewpoint
- Use relative (percentage, %) dimensions: width, height
- Use media-queries to apply different styles to large/small screens

Responsive web design (2)

- Use **@media** to apply css style only if a condition is met

```
@media only screen and (max-width: 500px) {  
  #div1 {  
    width: 100%;  
  }  
}
```

```
@media only screen and (min-width: 500px) {  
  ....  
}
```

```
@media only screen and (orientation: landscape) {  
  ...  
}
```

CSS neat typography: web fonts

- See Google web fonts: <https://fonts.google.com/>
- Usage example:

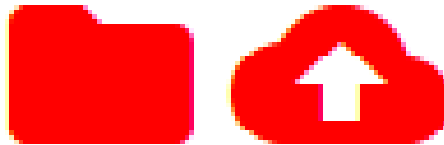
```
<link rel="stylesheet" type="text/css"
      href="https://fonts.googleapis.com/css?family=Jura">
<style>
  body {
    color: #cfd2da;
        font-family: 'Jura', sans-serif;
        font-size: 0.9rem;
  }
</style>
```

Neat CSS icons: google icons

```
<link rel="stylesheet"  
href="https://fonts.googleapis.com/icon?family=Material+Icons">
```

```
<i class="material-icons" style="font-size:48px;color:red">folder</i>
```

```
<i class="material-icons" style="font-size:48px;color:red"> cloud_upload</i>
```



Neat CSS icons: fontawesome

```
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css">
```

```
<i class="fa fa-car" style="font-size:60px;color:red;"></i>
```



Thank you!



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